# Jingyun Yang

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## Research Interests

Deep Learning, Computer Vision, Robotics

## Education

**University of Southern California** — Los Angeles, CA
Major in Computer Science · Minor in Mathematics · GPA: 4.0 / 4.0 — *August 2016 - May 2020*

## Experience

**USC Cognitive Learning for Vision and Robotics (CLVR) Lab** — Los Angeles, CA
*Research Assistant · Supervisor: Joseph J. Lim* — *August 2018 - May 2020*

- **Neural Program Synthesis from Visual Demonstrations:** Working on a deep learning model that synthesizes behavior programs for realistic videos where behavior segments are temporally extended.
- **Learning Coordinated Tasks from Rewards:** Designed, implemented, and tested a hierarchical reinforcement learning model that tackles complex tasks of multiple agents by learning composable primitive skills and coordination of the skills. Paper under review in ICLR 2020.
- **Video Prediction with Temporal Hierarchy:** Developed experiments and designed downstream planning algorithm for a model that learns to discover informative frames in a future video sequence and represent the video via its keyframes. Paper under review in ICLR 2020.

**USC Integrated Media Systems Center** — Los Angeles, CA
*Research Assistant · Supervisor: Yao-Yi Chiang* — *August 2016 - August 2018*

- **Neural Style Transfer:** Experimented with deep learning methods that transform an unusable map dataset with point annotations for texts to a usable dataset with bounding box annotations.
- **Timeseries Data Analytics:** Analyzed dynamic and large-scale transportation sensor data sent by buses in LA.

**Bloomberg L.P.** — New York, NY
*Software Engineering Intern · Supervisor: Luke Shen* — *May 2018 - August 2018*

- **Backend Software Development:** Migrated core functionalities of the Company Financials Core Algorithms Library to Spark in Scala.

**USC Information Sciences Institute** — Los Angeles, CA
*Research Assistant · Supervisor: Yolanda Gil* — *May 2017 - August 2017*

- **Data Visualization:** Created a module that automatically generates visualizations for either an arbitrary dataset or a group of datasets in the workflow system.

## Publications

1. Y. Lee, **J. Yang**, J. J. Lim, "Learning to Coordinate Manipulation Skills via Skill Behavior Diversification", Preprint, 2019.
2. K. Pertsch*, O. Rybkin*, **J. Yang**, K. Derpanis, J. J. Lim, K. Daniilidis, A. Jaegle, "Keyframing the Future: Discovering Temporal Hierarchy with Keyframe-Inpainter Prediction", ArXiv, 2019.

## Awards and Honors

- USC Provost's Undergraduate Research Fellowship — *August 2019*
- Tau Beta Pi the Engineering Honor Society — *April 2018*
- USC Academic Achievement Award — *August 2017*

## Teaching

- CSCI 170 – Dicrete Methods in Computer Science, Course Producer, USC — *Fall 2019, Spring 2019*
- CSCI 270 – Introduction to Algorithms and Theory of Computing, Course Producer, USC — *Fall 2018*